

Reflection on the McGill Course Advisor Agent Fix

The project is a single-agent OpenAI Agents SDK advising prototype using fake McGill-style fixtures. The eval failure was not that the tools lacked information; the tools already exposed course matches, prerequisites, and policy risks. The problem was how the live agent transformed that tool output into the structured CourseRecommendation object. Before the fix, the eval suite passed 4 of 7 cases. After the fix, it passed 7 of 7.

What failed

Three cases failed. In the COMP 251 case, the student had MATH 240 in progress but was missing COMP 250, so COMP 251 was not currently eligible. The agent noticed the missing prerequisite but still placed COMP 251 in recommended_courses. In the COMP 421 case, the course was missing COMP 251 and was also a 400-level course for a U1 student, yet the agent recommended it. In both cases, recommended_courses was being treated as a list of courses to discuss rather than courses the student could reasonably take now.

Before-fix logic: "Recommend courses only when they exist in the fake catalog.

Include prerequisite caveats, policy risks, and human approval needs..."

That instruction was too permissive because it allowed a course to be both recommended and labeled risky.

The third failure was different. For BIOL 999, the agent correctly returned no recommended courses and said the course was not found in the fake catalog, but the eval expected the exact phrase "not in the fake fixture catalog." That was a brittle wording test, not a true behavior failure.

What improved

The main fix clarified the meaning of a recommendation. The agent instructions now say that recommended_courses is only for courses the demo student can reasonably take now.

Courses with missing prerequisites, advisor-approval needs, unknown catalog status, completed or in-progress status, or high/medium policy risk must be discussed in summary, policy_risks, required_approvals, and next_steps instead.

After-fix logic: "Do not put a course in recommended_courses if it has missing prerequisites, needs advisor approval, is unknown to the fake catalog..."

After the fix, COMP 251 was no longer recommended; COMP 250 became the eligible next step. COMP 421 was also removed from recommended_courses, while its missing prerequisite and 400-level risk were preserved in the risk and approval fields.

The BIOL 999 eval was improved by changing expected_terms to ["BIOL 999", "fake catalog"], which checks the important behavior without requiring one exact sentence.

What is still risky

The biggest remaining risk is that the safety boundary is still mostly prompt-based. The tools detect missing prerequisites and policy risks, but no deterministic validator removes ineligible courses from recommended_courses before output. A stronger design would validate every recommended course against check_prerequisites and flag_policy_risk before returning the final answer.

The evals also still rely partly on string matching, and the guardrail is keyword-oriented, so indirect approval or overload requests may slip through.

Finally, because the catalog is fake and tiny, closest eligible recommendations such as COMP 250 for database interest are useful for testing but should not be mistaken for official advising.